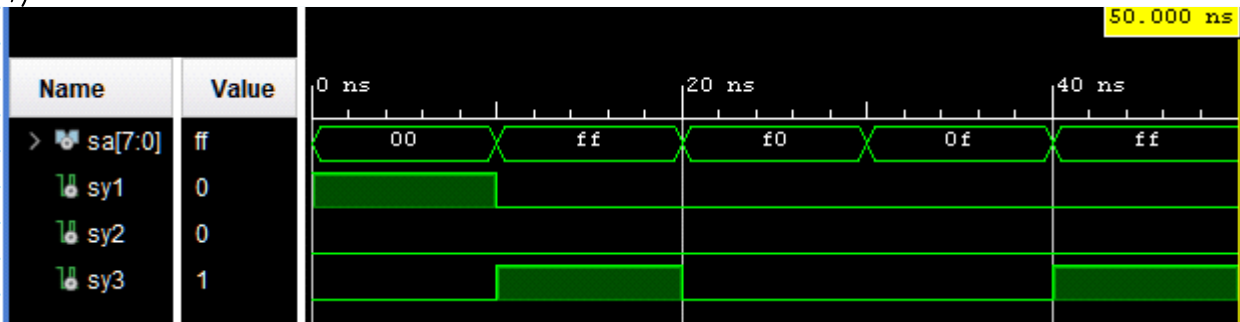


Homework #1

Wednesday, September 2, 2026 12:48 AM

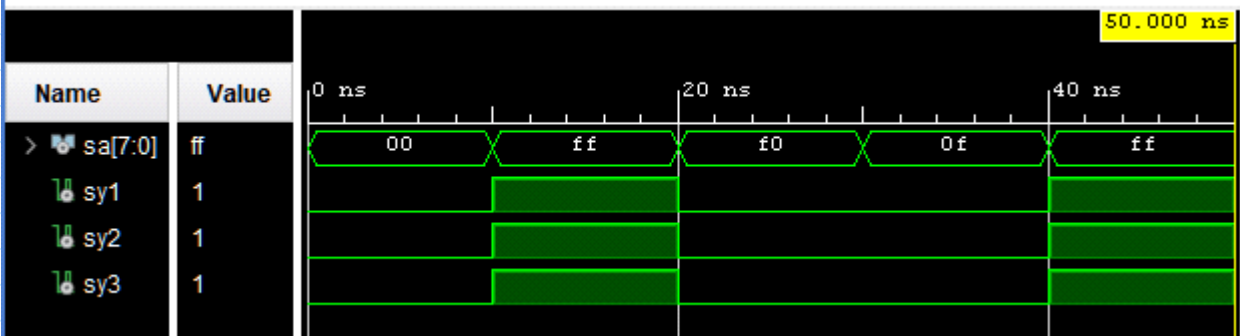
1)



System 3 is the only process that works like an 8 input AND gate

System 1 uses a signal instead of a variable causes `s_tmp` to not update when needed

System 2 never sets the variable `v_tmp` back to '1'



```
process (a)
variable s_tmp : std_logic := '1';
begin
    s_tmp := '1';
    for i in 0 to 7 loop
        s_tmp := s_tmp and a(i);
    end loop;
    y1 <= s_tmp;
end process;
```

```
process (a)
variable v_tmp : std_logic := '1';
begin
    v_tmp := '1';
    for i in 0 to 7 loop
        v_tmp := v_tmp and a(i);
    end loop;
    y2 <= v_tmp;
end process;
```

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.NUMERIC_STD.ALL;
```

```

entity XilinxIntro is
generic ( input_width : integer := 8 );
port ( a      : in std_logic_vector (input_width - 1 downto 0);
      y1, y2, y3 : out std_logic );
end XilinxIntro;

architecture rtl of XilinxIntro is
begin
-----
process (a)
variable s_tmp : std_logic := '1';
begin
  s_tmp := '1';
  for i in 0 to input_width - 1 loop
    s_tmp := s_tmp and a(i);
  end loop;
  y1 <= s_tmp;
end process;
-----
-----
process (a)
variable v_tmp : std_logic := '1';
begin
  v_tmp := '1';
  for i in 0 to input_width - 1 loop
    v_tmp := v_tmp and a(i);
  end loop;
  y2 <= v_tmp;
end process;
-----
-----
process (a)
-----
function f_and (a : in std_logic_vector (input_width - 1 downto 0)) return std_logic is
variable v_tmp : std_logic := '1';
begin
  for i in 0 to input_width - 1 loop
    v_tmp := v_tmp and a(i);
  end loop;
  return v_tmp;
end function f_and;
-----
begin
y3 <= f_and(a);
end process;
-----
end architecture;

```

```

library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.NUMERIC_STD.ALL;

entity XilinxIntro is
generic ( input_width : integer := 4 );

```

```

library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.NUMERIC_STD.ALL;

entity XilinxIntro is
generic ( input_width : integer := 4 );
port ( a      : in std_logic_vector (input_width - 1 downto 0);
      y1, y2, y3 : out std_logic );
end XilinxIntro;

```

